

Rishabh Sharma

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Indore, India

Professional Summary

AI/ML Engineer and production system architect with 3.5+ years of experience designing and delivering production-grade AI systems across Machine Learning, Agentic AI, GenAI, Data Science, and Deep Learning, including RAG, GraphRAG, multi-agent orchestration with LangGraph, LLM evaluation, and Prompt Engineering. Full-stack development background in Python and Java, with forward-deployed engineering experience building and embedding scalable enterprise AI solutions directly into client workflows. Skilled across DevOps and MLOps – from LLM fine-tuning to production ML pipelines on Azure MLOps and AWS SageMaker – delivering measurable business impact through computer vision, NLP, and IoT systems deployed at scale.

Experience

InfoBeans Technologies – Indore, India

AI/ML Engineer & Python Full Stack Developer

Aug 2024 – Present

- Built a hybrid **GraphRAG cognitive search system** over 1.5M+ documents on a **high-traffic production system** using Azure AI Search and Neo4j, cutting infrastructure costs **80%** and accelerating retrieval by 60%.
- Built a **HIPAA-compliant PII redaction system** (CRF + fine-tuned DeBERTa-v3), achieving **0.965 F1/0.3%** leakage vs. Presidio's 0.396 F1/66% leakage, deployed on Azure at 205ms/doc.
- Built a **document-to-Markdown parsing pipeline** using a fine-tuned DocLayNet detector with OCR and table extraction, achieving **95%+ structural accuracy** and 2× faster preprocessing than OCR-only pipelines.
- Led 5 developers building a **multi-agent audit orchestration system** (LangGraph, LangChain, Azure AI Foundry, Anthropic, Django, Azure DevOps CI/CD), architected to scale from 5,000 to **10M+ users**, cutting audit time by **75%**.
- Contributed to **large-scale system design** beyond AI workloads, including a full-stack **enterprise CRM platform** (FastAPI, React/Next.js, Kafka, Redis) and an **Enterprise Enrollment Portal** handling 16M peak requests across **400 auto-scaled containers**.

AI/ML Intern

Feb 2024 – Aug 2024

- Fine-tuned a YOLO model to detect PPE kits in night-vision footage for workplace safety monitoring, achieving **76% accuracy**.
- Promoted to full-time AI/ML Engineer after 6 months based on performance.

Software Engineering Intern (AI ML), InfoBeans Foundation – Indore, India

Nov 2022 – Jan 2024

- Learned and applied **full-stack development** using **Java** and the **MERN stack**, building real-world projects under mentorship.
- Worked on multiple **production-grade, AI-enabled full-stack web applications** using **Python, Java, MERN**, and **Azure**.

Projects

AI-Powered Enterprise CRM Platform – Python Full Stack

- Contributed across the stack (**FastAPI** backend, **React/Next.js** frontend, **PostgreSQL, Redis, Kafka**) on a multi-tenant enterprise CRM platform, building REST APIs and dashboards for lead, sales, and customer management workflows.
- Contributed to **LLM/RAG-powered** features (lead scoring, AI customer summaries, natural-language CRM queries) and platform infrastructure – Docker, Kubernetes, CI/CD (GitHub Actions), and observability (Grafana, OpenTelemetry).

Enterprise Enrollment Portal – Insurance & Payroll Platform

- Built a full-stack platform (**Django** backend, **React.js** web app, **React Native** mobile app) enabling organizations to enroll employees in group insurance and manage their complete payroll cycle through a ready-made payroll management system.
- Deployed on **Azure Container Apps** with horizontal autoscaling up to **400 concurrent containers** across multiple geographic regions, handling peak traffic of **16M requests** with full traceability/logging and a large-scale **PostgreSQL** layer backed by automated disaster-recovery backups.

AI-Powered Cognitive Search System (RAG + GraphRAG + OmniRAG) – CSA Group

- Built a hybrid **GraphRAG cognitive search system** over 1.5M+ multimodal documents using **Azure AI Search** and **Neo4j Knowledge Graphs**, cutting infrastructure costs **80%** and accelerating retrieval by 60%.

AI-Driven Audit Management System – CSA Group

- Led 5 developers building a **multi-agent audit orchestration system** (LangGraph, LangChain, Azure AI Foundry, Anthropic, Django, Azure DevOps CI/CD), architected to scale from 5,000 to **10M+ users**, cutting audit time by **75%**.

AI Interview Agent – Autonomous Candidate Screening Bot

- Built an **autonomous AI agent** using Python, Azure Virtual Machines, Playwright automation, ElevenLabs (TTS/STT), and Azure OpenAI that joins live candidate interviews, conducts real-time conversational screening with barge-in support, and tracks eye movement to assess engagement.
- Automatically evaluated candidate performance and delivered structured feedback to the hiring team, eliminating manual developer screening rounds entirely and saving **\$10,000/month** versus a purchased third-party service.

Temporal Graph-Based Kiosk Forecasting Pipeline

- Built a Python-based time-series pipeline that fetches client kiosk data every 15 days to forecast per-kiosk earnings and traffic, storing results in a **temporal knowledge graph in Neo4j** – converting a static schema into a time-indexed structure – reducing manual analysis effort to just **5%** of prior levels with measurable business impact within the first year.

Insurance Plan Advisor Chat Assistant

- Built a chat assistant plugged into an existing insurance portal to help users choose plans, calculate coverage, and get personalized recommendations and upsell suggestions – increasing customer satisfaction by **35%** and cutting enrollment completion time by **47%**.

AI-Based Tender & RFP Management System

- Built a full-stack AI-powered Tender & RFP management system (**Django** backend, **React** frontend) on Azure, with complete HLD/LLD system design and CI/CD pipelines, architected to scale to **20M+ users**.
- Automated document parsing, evaluation, and response generation using NLP, document extraction, and **Pinecone** vector search with OpenAI-based retrieval, generating real-time eligibility scores and risk insights; automated form filling and draft proposal generation, cutting turnaround from days to hours.

Face Recognition Attendance & Time Management System

- Built an end-to-end attendance portal and mobile app using **face recognition** for check-in/check-out, with a **FastAPI** (Python) backend and **SQL Server** for data management, deployed on in-house servers and used organization-wide.
- Built an integrated **Time Master** module logging working hours and cross-validating them against captured attendance data, with **React** (web portal) and **React Native** (mobile app) front-ends.

Technical Skills

Programming Languages: Python, SQL, Java, MERN (JavaScript)

AI & Machine Learning: Computer Vision, NLP, Deep Learning, Transfer Learning, Regression Models, LLMs, RAG, Multimodal RAG, LLM Evaluation, Time Series Analysis, OCR, Hyperparameter Tuning, Prompt Engineering, Agentic AI

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, FastAPI, Django, LangChain, LangGraph, Azure SDK

Data Analysis & Processing: Pandas, NumPy, SciPy, Matplotlib, Multiprocessing

Deployment & System Architecture: Docker, Kubernetes, Microservices, RabbitMQ, MLflow, Git, Redis, OpenTelemetry, CI/CD, AWS SageMaker, GitHub Actions, Azure DevOps Pipelines, Azure Service Bus, Ollama

Cloud Platforms: AWS, Microsoft Azure, Azure Blob Storage, Google Cloud Platform (GCP)

Databases: MongoDB, SQL-based Databases, Pinecone, Chroma, Supabase, Databricks

Tools: Git, Jupyter Notebooks, REST APIs, Jira, JMeter

Certifications

Oracle Certified GenAI Professional | Oracle Certified Vector Database and Search Professional | Oracle Certified Data Science Professional | AWS Cloud Practitioner Essentials | Microsoft Certified: AZ-400 (DevOps) | Microsoft Applied Skills: Azure AI Vision

Education

Maharaja Ranjeet Singh College, India

2019 – 2022

Bachelor of Science in Biotechnology, Life Sciences & Chemistry – CGPA: 8/10